

Research Statement

Continual Learning and Reinforcement Learning with Process Feedback

My research in continual learning began with a question about task order: why can changing the order of the same tasks alter what a model retains and how it adapts afterward? This question became the starting point for my first-author manuscript under review at IEEE TPAMI. The work led me to think of learning history as part of the learner’s current state, since it can shape later updates even when its effect is not yet visible in current performance. Later, during an internship on automated research workflows, I ran into another mismatch. A failed replication or diagnostic check could improve the next decision, while a sequence that looked productive could leave the final evidence no stronger. That experience led to a separate, still developing interest in reinforcement learning with process feedback. I treat the two directions separately, but both ask what a current score leaves out about what a learner or agent will do next.

1 How does learning history change what a model can learn next?

In my manuscript, *Stability–Plasticity Theory for Continual Learning under Drifting Task Optima*, I study a sequential linear learner whose past prediction errors shape a history-dependent update metric. The analysis gives a tracking guarantee for drifting task optima and relates retention to current-task error and movement between optima. A controlled three-task construction fixes the pairwise distances among task optima and changes only their order. Different orders produce different learned metrics and retention behavior; when metric learning is disabled, the order effect disappears. The model is deliberately simplified. It does not establish the same mechanism in deep networks, but it isolates one way in which history can become part of the learner state. It also made me less willing to treat current loss or predictions as a complete description of readiness for future learning.

Related work gives important partial answers. Single-turn anchors can counter obsolete reasoning traces within a multi-turn episode, and Complementary RL updates an external experience bank alongside the actor [1, 2]. In model adaptation and continual learning, gradient reconstruction and gradient-matching coresets show how limited stored information can approximate past gradients or support rehearsal [3, 4]. These methods act on different carriers and timescales, which makes the carrier itself part of the question.

The narrower question I want to study is whether, after capacity, training budget, and current behavior are closely matched, any history-induced learner state still predicts how a model will respond to the same subsequent data. A first test would train models through different task sequences, match their current losses and predictions, and then compare later adaptation and forgetting. Candidate state variables include update geometry and representation structure; learner-relative memory utility provides one setting in which to test them. If a candidate state is causal, resetting or exchanging it should change later behavior in the expected direction. With foundation models, the same question extends to experience that differs in modality, reliability, or level of abstraction, under fixed storage and update budgets. If current-gradient coverage or representation similarity already explains the result, I would set aside the stronger history-state account.

2 What should process feedback reward?

My second direction began as a measurement problem. In long research trajectories, an action can matter because it rules out a tempting explanation, verifies a prerequisite, or shows that the available evidence is too weak to continue. None of these actions must raise the immediate task score. At the same time, polished reasoning or repeated local improvements can look like progress without changing which conclusion is justified.

Process feedback itself is not an untouched problem. Process advantage verifiers define progress as the change in future answer success under a separate prover policy and improve search and online reinforcement learning on verifiable reasoning tasks [5]. PRIME learns dense implicit process rewards online from rollouts and outcome labels [6], while turn-level rollout values have supported iterative policy improvement in ALFWorld [7]. Astar provides a different example: it learns evolution proposals from industrial iteration histories and periodically recalibrates its surrogate reward model using real executions [8]. My question begins in settings where a single correctness or success label is inadequate: what should a process target represent when final utility combines task outcome, evidence validity, and cost, and the agent must decide whether to explore, repeat, switch, or stop?

Drawing on the Bayesian selection framework of Hay et al. [9], I would first construct finite experiment-selection environments where reference action values can be computed. An agent chooses among noisy, costly experiments before selecting a candidate or stopping. This makes it possible to compare immediate improvement, eventual success probability, information gain, and counterfactual decision value for repeating, switching, and stopping. If an exact predictor of success probability still ranks evidence-seeking actions incorrectly, the problem is target specification rather than estimation. I would then audit learned signals on matched trajectories, varying whether a step supplies valid evidence, resolves a prerequisite, or merely accompanies a good outcome. The important failures are false progress and the punishment of informative failure.

Only signals that survive these tests should be used for trajectory selection, best-of- N , or policy optimization. I would compare outcome-only, process-only, and combined objectives under matched settings and total budgets, with final utility measured by an evaluator that did not define the process reward. If optimization raises the learned process score but not independently verified outcomes at the same cost, I would not count it as improvement.

Research Perspective

Continual learning is my established research direction; process-feedback reinforcement learning is an emerging one shaped by practical experience and reading. Across both, I start from a concrete mismatch, turn it into a precise question, and test it against competing explanations. In the long term, I hope to build learning systems that can use the past without being trapped by it, and that finish a difficult task better prepared for the next one.

References

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